CSC 2050 – Assignment 2: Dynamic Array

The C language is poor when it comes to high-level data structures compared to other languages. However, by providing direct access to low-level resources such as memory management, developers can build their own data structures and customize them for their needs.

For this assignment, you need to create a dynamic array. Your custom dynamic array should behave like an ArrayList in Java or like a Vector in C++. That is, a user of your dynamic array should be able to add elements in the array without any size limitations. However, the data type that your array will hold will be static, meaning that your array will hold only integers, characters, float, doubles and other types at once. Combining multiple data types in one array shouldn't be allowed.

There are two problems to tackle on this assignment. How to represent a dynamic array in code, and what functionality you should provide. The provided dynamicArray.c file provide some answers to these questions. The DynamicArray struct is already defined for you. It contains a pointer to the data the array will hold, and two variables for the size and the capacity. You should use that struct to implement the functions. All the functionality you should implement for the dynamic array is also provided. Read the comments before every function to understand its functionality, and then implement the function. Note that some of the pre-conditions for the functions need to be tested for errors before you move on with the implementation. For example, if a function has a pre-condition that the dynamic array is not null, you need to check if the array is null and exit the function if it is providing an error message.

Your task is to complete the functions for the dynamic array functionality, and create test cases in the main function to test those functions.

Make sure that your code compiles without errors or warnings or memory leaks, and runs on the Linux servers.

Make sure that your code is documented enough. 30% of the grade is for documentation.

Grading:
Each dynamic array function takes 5 points (50 total), 30 points for the test cases in the main function, and 20 points for the documentation and the style.

Submission:
After you complete the assignment, submit it on CANVAS.

Extra Credit:
Once you're done and have submitted everything, there's one last thing that takes 5-10 minutes! As Silas announced in class, he is running research on how students use AI. Please use this link below to either complete the research study or do the alternative reflection activity. Participation in the research is voluntary and confidential to the course instructors, but you must either complete the survey or the reflection activity and explicitly decline participation to receive credit.

https://stevens.qualtrics.com/jfe/form/SV_42w3fWc8opmT0l8?Course=CSC2050-07&Assignment=2